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$$z = x_1 + 2x_2 + 4x_3 + x_4 \rightarrow \max$$

$$\forall i \in \mathbb{N} \hookrightarrow x_i \geq 0$$

$$\begin{cases} x_1 + x_2 + x_3 \leq 10 \\ x_2 + 2x_3 + x_4 = 6 \\ x_1 + x_4 \geq 2 \end{cases} \Leftrightarrow \begin{cases} x_1 + x_2 + x_3 + x_5 = 10 \\ x_2 + 2x_3 + x_4 = 6 \\ x_1 + x_4 - x_6 = 2 \end{cases}$$

$$\Leftrightarrow \left[\begin{array}{cccccc|c} 1 & 1 & 1 & 0 & 1 & 0 & 10 \\ 0 & 1 & 2 & 1 & 0 & 0 & 6 \\ 1 & 0 & 0 & 1 & 0 & -1 & 2 \end{array} \right]$$

$$\Leftrightarrow \left[\begin{array}{cccccc|c} 0 & 1 & 1 & -1 & 1 & 1 & 8 \\ 0 & 1 & 2 & 1 & 0 & 0 & 6 \\ 1 & 0 & 0 & 1 & 0 & -1 & 2 \end{array} \right]$$

$$\Leftrightarrow \left[\begin{array}{cccccc|c} 0 & 0 & -1 & -2 & 1 & 1 & 2 \\ 0 & 1 & 2 & 1 & 0 & 0 & 6 \\ 1 & 0 & 0 & 1 & 0 & -1 & 2 \end{array} \right]$$

$$\Leftrightarrow \left[\begin{array}{cccccc|c} 1 & 0 & 0 & 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & 1 & 0 & 0 & 6 \\ 0 & 0 & -1 & -2 & 1 & 1 & 2 \end{array} \right]$$

	<u>1</u>	<u>2</u>	4	1	<u>0</u>	0	
	x_1	x_2	x_3	x_4	x_5	x_6	
x_1	1	0	0	1	0	-1	2
x_2	0	1	2	1	0	0	6
x_5	0	0	-1	-2	1	1	2
	0	0	0	-2	0	1	14

	<u>1</u>	<u>2</u>	<u>4</u>	1	0	<u>0</u>	
	x_1	x_2	x_3	x_4	x_5	x_6	
x_1	1	0	-1	-1	1	0	4
x_2	0	1	2	1	0	0	6
x_6	0	0	-1	-2	1	1	2
	0	0	1	0	-1	0	16

	<u>1</u>	2	<u>4</u>	1	0	<u>0</u>	
	x_1	x_2	x_3	x_4	x_5	x_6	
x_1	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	0	7
x_3	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	0	3
x_6	0	$\frac{1}{2}$	0	$-\frac{3}{2}$	1	1	5
	0	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-1	0	19

$$\begin{cases} x_1 = 7 \\ x_2 = 0 \\ x_3 = 3 \\ x_4 = 0 \\ x_5 = 0 \\ x_6 = 5 \end{cases}$$

$$z = x_1 + 2x_2 + 4x_3 + x_4 \rightarrow \max$$

$$\forall i \in \mathbb{N} \hookrightarrow x_i \geq 0$$

$$\begin{cases} x_1 + x_2 + x_3 \leq 10 \\ x_2 + 2x_3 + x_4 = 6 \\ x_1 + x_4 \geq 2 \end{cases}$$

$$\Leftrightarrow \begin{cases} x_1 + x_2 + x_3 + x_5 = 10 \\ x_2 + 2x_3 + x_4 = 6 \\ x_1 + x_4 - x_6 = 2 \end{cases}$$

$$\Leftrightarrow \left[\begin{array}{cccccc|c} 1 & 1 & 1 & 0 & 1 & 0 & 10 \\ 0 & 1 & 2 & 1 & 0 & 0 & 6 \\ 1 & 0 & 0 & 1 & 0 & -1 & 2 \end{array} \right]$$

$$\left[\begin{array}{cccccccc|c} 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 2 & 1 & 0 & 0 & 1 & 0 & 6 \\ 1 & 0 & 0 & 1 & 0 & -1 & 0 & 1 & 2 \end{array} \right]$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	
x_5	1	1	1	0	1	0	0	0	10
x_7	0	1	2	1	0	0	1	0	6
x_8	1	0	0	1	0	-1	0	1	2
	1	2	4	1	0	0	0	0	0

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	
x_3	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	0	$\frac{1}{2}$	0	3
x_5	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	0	$-\frac{1}{2}$	0	7
x_8	1	0	0	1	0	-1	0	1	2
	1	0	0	-1	0	0	-2	0	12

	<u>1</u>	2	<u>4</u>	1	<u>0</u>	0	0	0	
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	
x_1	1	0	0	1	0	-1	0	1	2
x_3	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	0	$\frac{1}{2}$	0	3
x_5	0	$\frac{1}{2}$	0	$-\frac{3}{2}$	1	1	$-\frac{1}{2}$	-1	5
	0	0	0	-2	0	1	-2	-1	14

	<u>1</u>	2	<u>4</u>	1	0	<u>0</u>	0	0	
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	
x_1	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	1	0	$-\frac{1}{2}$	0	7
x_3	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0	0	$\frac{1}{2}$	0	3
x_6	0	$\frac{1}{2}$	0	$-\frac{3}{2}$	1	1	$-\frac{1}{2}$	-1	5
	0	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-1	0	$-\frac{3}{2}$	0	19

$$\begin{cases} x_1 = 7 \\ x_2 = 0 \\ x_3 = 3 \\ x_4 = 0 \\ x_5 = 0 \\ x_6 = 5 \\ x_7 = 0 \\ x_8 = 0 \end{cases}$$